Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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50(2.4,14.6) **CEITEC – Central European Institute of Technology**

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PROGRESS REPORT

for the TAC Meeting

Date of the Meeting: .....

I. GENERAL INFORMATION

PhD candidate: .....

The number of semesters completed: .....

PhD programme: .....

Supervisor: .....

Project title: .....

II. DESCRIPTION OF THE PROGRESS ON THE DISSERTATION PROJECT

A concise description of the activities and progress during the reported period - since the previous TAC meeting (ca 1 page).

[topsep=0.0pt, itemsep=-1ex]What have you been working on, and which milestones have you achieved? Have you reached the proposed goals? If not, why? Which modifications in terms of research questions, methodological approach, etc. have been made? What are the outcomes of your activity – which findings and results have been made? A concise overview of activities related to the fulfillment of study requirements – active participation at conferences, the internship completed, etc.

III. DESCRIPTION OF RESEARCH AND MILESTONES PLANNED FOR THE FOLLOWING PERIOD

A brief description of the plans – until the next TAC meeting/completion of the PhD study (ca 1/2 page).

[topsep=0.0pt, itemsep=-1ex]Specify the working steps and achievements planned. Specify the plans concerning attending conferences, completing an internship, and publishing the results of your research.

This report should be submitted two weeks before the TAC Meeting to all TAC members.